Improving the prediction of high-risk transport refusals

A field checklist derived from linked prehospital records

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Transport refusal

After treatment, any patient may refuse to be transported to hospital.

MONTREAL AND LAVAL · 2015–2022

8.5%

of 9-1-1 calls ended with the patient refusing transport

5.41%

of linked refusals were followed by hospitalization or death within 72 hours

WHAT HAPPENS ON SCENE

STEP 01

The patient refuses. The crew confirms capacity and explains the risks.

STEP 02 · REFUS À RISQUE

The crew checks the refusal against **19 high-risk criteria**

ANY CRITERION PRESENT

The crew must call clinical support before leaving the scene

NO CRITERION

No call

STEP 03 · EVERY REFUSAL

Recorded on audio and signed. The patient stays home.

A valid refusal should be respected. The task is to recognize the minority of refusals likely to end badly.

The *Refus à risque* criteria

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1	Under 14, no responsible adult on scene	2	Under 14 with any other risk criterion
3	Altered level of consciousness (V, P or U on AVPU)	4	New neurological deficit
5	Confusion, acute or chronic without a responsible adult	6	Breathing difficulty, reported or observed: SpO_2 < 92%, cyanosis, retractions, wheezing
7	Non-traumatic pain under 48 h: headache, abdominal, back	8	Chest pain or palpitations
9	Pregnancy with abdominal pain, vaginal bleeding or syncope	10	Active hemorrhage
11	Hypoglycemia below 4 mmol/L	12	Intervention under the symptom-relief programme (PSS)
13	Pulse below 50 or above 120 per minute	14	Allergic reaction
15	Syncope or convulsions	16	Systolic BP < 100 or > 200 mm Hg, or diastolic > 100, interpreted according to age
17	Head trauma with signs or symptoms	18	Any trauma with a positive EQTPT score
19	Any trauma with a Glasgow Coma Scale of 14 or below		

Expert opinion. Not yet evaluated against what happened to the patient afterwards.

Objectives and approach

Models were built using data from January 2015 to June 2022.
Models were evaluated using data from June 2022 to December 2022.

01 · Evaluate

How well does the current protocol perform?

19 criteria on the runsheet

↓

18 rebuilt from recorded variables

↓

Any criterion present → **high risk**

02 · Build

A better checklist for paramedics

LASSO: recorded variables → **83 predictors**

↓

Clinical review

↓

13 items, refit as a logistic regression

03 · Benchmark

A ceiling for decision support

XGBoost: gradient-boosted decision trees

↓

The same 83 predictors

↓

Does a flexible model do better?

Data and cohort

Urgences-santé, Montreal and Laval · January 2015 to December 2022

THREE LINKED DATA SOURCES

1. **Computer-aided dispatch**
   the 9-1-1 call and its closure
2. **Paramedic runsheet**
   what the crew recorded on scene
3. **Provincial health records**
   hospitalization or death within 72 hours

COHORT

2,060,005
single-patient calls

→

175,260
refusals

→

94,825
linked to both sources

5,130 (5.41%) hospitalized or died within 72 hours of the call

Performance of the current protocol

Refus à risque criteria applied to refusals from June to December 2022

Refusals flagged	75.8%
Sensitivity	77.7%
Specificity	24.4%
Positive predictive value	6.7%
Negative predictive value	94.0%

Most alarms were false, and about one in five patients who were later hospitalized or died was missed.

The 13-item checklist

Odds ratios from the refit logistic regression.

VITAL SIGNS MEASURED ON SCENE	
Tachycardia · heart rate > 100/min	**1.81**
Tachypnea · respiratory rate > 22/min	**1.51**
Hypoxia · SpO_2 < 94%	**1.88**
Hypertension · systolic BP > 180 mm Hg †	**0.71**
PATIENT BACKGROUND	
Older age · per 10 years	**1.39**
Medical history or current medication	**2.24**
Alcohol or drug intoxication †	**0.53**
PRESENTING COMPLAINT AT DISPATCH	
Unconscious or fainting †	**0.79**
Breathing difficulty	**1.35**
Cardiac problem or implanted defibrillator †	**0.63**
Signs of stroke or transient ischemic attack	**1.44**
Psychiatric presentation or suicide attempt	**1.63**
GENERAL APPEARANCE	
Abnormal general appearance or skin colour	**1.33**

† Kept on clinical grounds despite an odds ratio below 1: signs that are dangerous to miss.

Comparison of the three rules

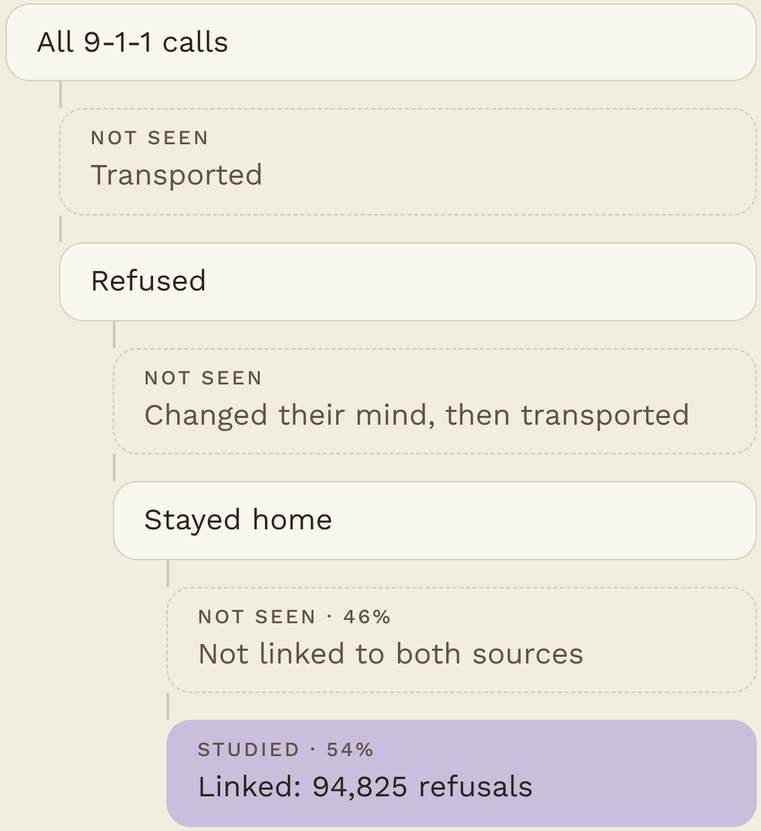
Thresholds set to match the protocol's sensitivity.

	Current protocol	13-item checklist	XGBoost
Refusals flagged	75.8%	46.3%	40.0%
Sensitivity	77.7%	83.7%	80.5%
Specificity	24.4%	56.4%	62.8%
Positive predictive value	6.7%	11.8%	13.0%
AUROC	0.619	0.758	0.783

For every 1,000 refusals, the checklist triggered about **300 fewer calls** to clinical support and missed about **4 fewer patients** who were hospitalized or died.

Limitations

What the data can and cannot see



Patients who refused transport are a selected group. Danger signs that usually lead to transport are rare among them. This does not mean the signs are safe.

Patients who changed their mind were recorded as transports. The protocol's true performance may be better than reported.

Half of refusals could not be linked. Patients who could not give a health insurance number may be the ones at higher risk.

Three takeaways

01 Under the current protocol, most refusals triggered a call to clinical support. Some patients who were later hospitalized or died were still missed.

02 A 13-item checklist did better on both counts. For every 1,000 refusals: about 300 fewer calls and about 4 fewer missed patients.

03 The checklist is not ready for use yet. It still needs review by clinical experts and testing in the field.

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